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REQPILOT AI: A FULL-STACK MULTI-AGENT PLATFORM FOR INTELLIGENT REQUIREMENTS ANALYSIS AND PROJECT PLANNING

Full Stack Development Project Report

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Modern software engineering relies fundamentally on the accuracy, clarity, and completeness of its initial requirements engineering phase. In software development life cycles (SDLC), errors introduced during requirements elicitation and specification represent the single most expensive category of software defects to rectify. Research consistently indicates that correcting an erroneous or ambiguous requirement during post-deployment maintenance costs up to one hundred times more than resolving it during the initial inception phase [1], [2]. Despite decades of process formalization, requirements in practice are routinely gathered from non-technical business stakeholders, product owners, and domain experts through unstructured text documents, interview transcripts, emails, and conversational meeting notes.

These raw requirement artifacts typically suffer from acute qualitative deficiencies: high ambiguity, passive linguistic voice, omitted acceptance boundaries, conflicting business rules, and a pervasive lack of traceability to technical architecture and testing infrastructure [3]. Requirements engineers and project managers are consequently burdened with labor-intensive, repetitive manual operations. They must painstakingly comb through narrative documentation, extract functional and non-functional requirements, detect contradictions, formulate clarification questions, synthesize modular system architectures, estimate implementation effort, decompose tasks into sprint backlogs, and maintain bidirectional traceability links across evolving project artifacts [4].

The emergence of generative artificial intelligence and Large Language Models (LLMs) has demonstrated transformative potential for natural language processing. However, deploying a naive, monolithic LLM prompt to analyze requirements introduces severe engineering hazards: stochastic hallucination of phantom technical constraints, unpredictable schema drift, absence of verifiable evidence grounding, and the complete inability to track dependency blast-radii when requirements evolve [5]. To overcome these fundamental vulnerabilities, this project introduces ReqPilot AI, an end-to-end full-stack platform that pairs multi-agent cognitive reasoning with deterministic engineering constraints, interactive human-in-the-loop review, and automated project planning intelligence.

1.2 DOMAIN OVERVIEW

ReqPilot AI operates at the intersection of three major technical domains: Full-Stack Web Engineering, Requirements Engineering (RE), and Multi-Agent Artificial Intelligence Systems [6]. Within the domain of full-stack engineering, the platform delivers a modern, high-performance web interface designed for project managers, business analysts, software architects, and agile developers. Built using React 19, TypeScript, Vite, and Tailwind CSS on the frontend, and FastAPI, Python 3.11, Uvicorn, and SQLAlchemy 2.0 Async ORM on the backend, the application provides a production-grade modular workspace that manages the entire lifecycle of software engineering specifications.

Within the requirements engineering domain, ReqPilot AI addresses the complete spectrum of standard ISO/IEC/IEEE 29148 requirements characteristics: completeness, consistency, clarity, testability, atomicity, feasibility, and traceability [7]. The platform moves beyond static document storage by transforming raw textual specifications into a connected, relational knowledge graph comprising projects, requirements, quality issues, clarification questions, architecture modules, epics, sprint tasks, and verification test cases.

Within the domain of artificial intelligence, ReqPilot AI implements an ensemble of fifteen specialized, cooperative agents orchestrated under a deterministic workflow engine. Rather than relying on black-box text generation, the platform combines heuristic pattern matching, regular expression extraction, lexical similarity matching, topological graph sorting, and LLM-assisted synthesis (via Google Gemini 1.5 Flash, Groq LLaMA-3.3-70B, or OpenAI GPT-4o-mini). This multi-tier architecture ensures that analytical insights are traceable, structurally auditable, and grounded in verifiable source document excerpts where direct textual evidence exists.

1.3 PROBLEM STATEMENT

The central problem addressed by ReqPilot AI is the absence of an integrated, automated, and mathematically grounded full-stack platform capable of converting unstructured stakeholder specifications into validated, architecturally decomposed, traceable, and change-aware project plans [8]. Traditional software engineering workflows suffer from the following critical failure modes:

1. Severe Ambiguity and Sub-optimal Quality: Requirements frequently utilize non-verifiable adjectives (e.g., 'fast', 'secure', 'user-friendly', 'scalable') without specifying quantitative service level agreements (SLAs) or empirical acceptance criteria.

2. Cross-Document Business Rule Conflicts: Large specifications written by disjoint stakeholder groups regularly contain mutually exclusive constraints (e.g., permitting unrestricted cancellation while simultaneously requiring a strict 24-hour non-refundable cutoff).

3. Disconnected Architecture and Task Planning: Requirements exist in isolation from engineering decomposition. System architects and scrum masters must manually guess module allocations, work breakdowns, and sprint allocations without automated structural guidance.

4. Absence of Change Impact Blast-Radius Analytics: When a requirement is modified mid-development, teams have no automated mechanism to determine which modules, development tasks, test cases, and delivery milestones are invalidated.

5. Opaque AI Automation: Conventional generative AI tools generate unverified requirements that lack source line grounding, hallucinate non-existent features, and exclude humans from the approval loop.

1.4 MOTIVATION

Empirical software engineering literature has consistently documented that a substantial fraction of software development projects experience budget overruns, timeline delays, or outright cancellation, with flawed, ambiguous, and shifting requirements identified as primary contributing factors [1], [7], [9]. While modern developer tooling has heavily automated continuous integration, containerized deployment, and code generation, the upstream requirements engineering phase remains manual and frequently fragmented across generic word processors and disconnected spreadsheet trackers.

A modern full-stack technological solution is urgently required to bridge this gap. By building a dedicated web platform that integrates automated document parsing, multi-dimensional quality scoring, conflict detection, architectural graph modeling, Directed Acyclic Graph (DAG) task scheduling, and change-impact blast-radius calculation, engineering teams can substantially mitigate weeks of administrative overhead. Furthermore, establishing a transparent human-in-the-loop review mechanism ensures that engineers retain authoritative control over requirement approvals and architectural commitments.

1.5 OBJECTIVES

The primary objective of ReqPilot AI is to deliver a functional, integrated full-stack software requirements engineering and project planning platform. The specific, measurable objectives achieved by the system include:

1. Multi-Format Document Ingestion: Ingest unstructured software specifications from PDF, Microsoft Word (.docx), and plaintext (.txt) formats, performing automated structural cleaning, normalization, and semantic chunking.

2. Structured Requirement Extraction: Automatically extract and classify functional requirements (FR) and non-functional requirements (NFR), capturing explicit structural attributes including Actor, Action, Target Object, Preconditions, and Expected Outcome.

3. 7-Dimensional Transparent Quality Auditing: Evaluate every requirement against an ISO/IEC/IEEE 29148 aligned mathematical framework spanning Completeness (20%), Clarity (15%), Atomicity (15%), Consistency (20%), Testability (15%), Feasibility (10%), and Traceability (5%).

4. Automated Ambiguity, Conflict, and Duplicate Detection: Identify subjective, non-measurable linguistic phrases, detect cross-requirement logical contradictions, and compute lexical similarity to mitigate duplicated specifications.

5. Clarification Question Generation: Formulate targeted, multiple-choice clarification inquiries with engineering rationales for low-scoring requirements, allowing human stakeholders to disambiguate specifications.

6. Automated Architecture Module Synthesis: Group extracted requirements into logical microservice and gateway components, generating inter-module dependency relations, API endpoints, and canvas coordinates.

7. Agile Task Decomposition and DAG Scheduling: Decompose requirements into development tasks with estimated hours and story points, applying NetworkX topological sorting to ensure cycle-free sprint execution order.

8. Verification Test Case Generation: Automatically generate integration, performance, and unit test cases linked directly to acceptance criteria for every functional requirement.

9. Bidirectional Traceability Matrix: Construct an end-to-end traceability matrix linking Source Excerpt -> Requirement -> Architecture Module -> Task -> Test Case.

10. Change Impact Blast-Radius Analysis: Simulate the ripple effect of requirement mutations across downstream modules, sprint tasks, and verification tests, computing schedule impact and risk severity.

1.6 SCOPE OF THE PROJECT

The operational scope of ReqPilot AI encompasses the complete early-to-mid software engineering lifecycle, specifically targeting the transition from raw stakeholder elicitation to sprint backlog execution. The platform supports project creation, multi-document parsing, interactive requirement filtering, automated quality audits, interactive clarification answering, architecture visualization, sprint task allocation, team skill matching, test case inspection, change-impact simulation, and consolidated engineering report generation [10].

The project boundaries are clearly demarcated: ReqPilot AI does not execute the generated source code, deploy cloud infrastructure, or directly integrate with live banking payment gateways. Instead, it provides the complete specification, orchestration, and planning metadata necessary for developers and architects to build systems accurately. The platform currently integrates an asynchronous SQLite persistence engine (with seamless migration capability to PostgreSQL and pgvector) and supports hybrid LLM execution across Gemini, Groq, and local deterministic fallbacks.

1.7 TARGET USERS

ReqPilot AI is purpose-built to support four primary categories of software engineering stakeholders:

1. Requirements Engineers and Business Analysts: Benefit from automated document parsing, ambiguity highlighting, 7-D quality breakdowns, and auto-generated clarification questionnaires that streamline client elicitation sessions.

2. Software Architects: Leverage automated architecture synthesis to inspect component boundaries, communication protocols (REST, WebSockets, Pub/Sub), database mappings, and inter-module dependencies.

3. Project Managers and Scrum Masters: Utilize automated task decomposition, story-point estimation, critical-path DAG scheduling, team capacity matching, and change-impact blast-radius analytics to prevent sprint disruption.

4. Quality Assurance (QA) and Test Engineers: Utilize auto-generated verification test suites, acceptance criteria mappings, and end-to-end traceability links to achieve comprehensive verification test coverage against approved specifications.

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CHAPTER 2

EXISTING SYSTEM AND DOMAIN STUDY

2.1 EXISTING SYSTEM

In traditional software engineering practice, requirements analysis and project management are performed across disparate, disconnected commercial tools. Requirements elicitation typically begins in unstructured office document software such as Microsoft Word, Google Docs, or Confluence. Stakeholders draft Software Requirements Specifications (SRS) based on IEEE Std 830-1998 templates, containing narrative descriptions of system behaviors, business rules, and interface constraints [11].

Once drafted, requirements are manually reviewed through peer walkthroughs, inspection checklists, and steering committee meetings. Following approval, project managers manually transcribe these specifications into issue-tracking and backlog-management platforms such as Jira, Azure DevOps, Asana, or Trello. System architects simultaneously utilize standalone modeling tools (e.g., Lucidchart, Enterprise Architect, PlantUML) to draft structural diagrams, while quality assurance engineers maintain test cases in dedicated test repositories (e.g., TestRail, Zephyr) [12]. In recent years, generative AI chat assistants such as ChatGPT, Claude, and GitHub Copilot Chat have been informally adopted by engineers for one-off brainstorming, drafting user stories, or summarizing documents.

2.2 LIMITATIONS OF EXISTING SYSTEM

Despite the maturity of individual project management and modeling utilities, existing industry approaches suffer from profound systemic limitations when handling requirements engineering:

1. Severe Semantic Fragmentation: Requirements, architecture models, sprint backlogs, and test cases live in isolated data silos. There is no automated data fabric linking a change in a Word document to a Jira task or a TestRail test script.

2. Inability to Detect Latent Ambiguities: Standard word processors provide grammatical and orthographic checks but lack specialized requirements-engineering semantic validation. Subjective adjectives like 'the system must scale smoothly' pass through manual walkthroughs unnoticed, resulting in costly re-engineering during performance testing [13].

3. Manual, Error-Prone Conflict Identification: In large enterprise specifications spanning dozens of pages, detecting contradictory business logic (such as conflicting session cancellation cutoffs) requires exhaustive manual cross-referencing, which human reviewers routinely miss.

4. Arbitrary and Uncalibrated Effort Estimation: Agile story pointing and hour estimates are frequently assigned during hurried planning poker sessions without reference to linguistic complexity, technical risk, or historical task structures.

5. Pervasive Traceability Decay: While compliance standards demand bidirectional traceability matrices, maintaining links between hundreds of requirements, code modules, and tests in static spreadsheets is exceedingly tedious. In practice, traceability matrices decay within weeks of project kick-off.

6. Catastrophic Change Impact Blindness: When a client alters a business rule midway through a sprint, teams lack an automated mechanism to compute the blast radius. Developers frequently modify backend code while forgetting to update authorization gateways, resulting in silent regressions and missed sprint deadlines [14].

7. Hallucination Hazards of Generic Chatbots: Generic chat assistants lack software engineering schemas, produce non-deterministic outputs, invent phantom API endpoints, and lack grounding in source document line numbers.

2.3 EXISTING TECHNOLOGIES / APPROACHES

Academic and industrial research in automated requirements engineering has explored several distinct paradigms over the past two decades:

1. Rule-Based and Regex Parsers: Early natural language processing (NLP) approaches utilized part-of-speech (POS) tagging and grammatical parsers (e.g., Stanford CoreNLP, spaCy) to identify modal verbs ('shall', 'must') and extract subject-verb-object tuples. While fully explainable, rule-based systems are brittle, failing when requirements deviate from strict grammatical structures [15].

2. Machine Learning Classifiers: Supervised machine learning algorithms (e.g., Naive Bayes, Support Vector Machines, Random Forests, BERT) have been deployed to classify requirements into functional and non-functional categories. While effective for classification, these models do not synthesize architectural designs, generate test cases, or manage project plans.

3. Ontological and Graph-Based Models: Semantic Web ontologies and Knowledge Graphs represent requirements as entity networks. Graph traversals enable dependency analysis but require immense upfront modeling effort and lack natural language generation capabilities [16].

4. Prompt-Based Large Language Models: Recent commercial tools utilize LLM zero-shot prompts to summarize text into user stories. However, monolithic prompts suffer from attention dilution on large documents, lack deterministic validation, and lack structural ACID database persistence.

2.4 PROPOSED IMPROVEMENT

ReqPilot AI introduces a cohesive architectural approach by establishing an integrated Multi-Agent Cooperative Engineering Fabric. Rather than relying solely on brittle rule-based parsers or unconstrained stochastic LLM generators, ReqPilot AI couples structured natural language reasoning with deterministic validation rules within a schema-validated execution pipeline [17].

The proposed platform provides five transformative improvements: (1) Deterministic Ingestion and Normalization that converts complex documents into semantic chunks with exact line numbers and page anchors; (2) A 15-Agent Multi-Tier Hierarchy where each agent possesses an atomic, specialized engineering responsibility; (3) A Transparent 7-Dimensional Quality Scoring Engine with explicit, configurable weights aligned with ISO/IEC/IEEE 29148; (4) Topological DAG Graph Scheduling that automatically detects cyclic dependencies and computes optimal execution orders; and (5) An Automated Change-Impact Blast-Radius Engine that traverses the relational graph to identify affected artifacts with zero human guesswork.

2.5 NOVELTY AND KEY CONTRIBUTION

The novelty of ReqPilot AI lies not merely in calling LLM APIs, but in its unique architectural synthesis of cognitive AI, relational persistence, and deterministic software

engineering controls. Table 2.1 contrasts existing industry tools with the implemented ReqPilot AI platform.

Table 2.1 Comparative analysis of existing approaches vs. ReqPilot AI

Feature / Capability	Jira / Azure DevOps	Confluence / Word	ChatGPT / Generic AI	ReqPilot AI (Proposed)
Automated Requirement Extraction	No (Manual Entry)	No (Manual Drafting)	Partial (Unstructured Text)	Yes (Evidence-Grounded Parsing)
7-D ISO Quality Auditing	No	No	No (Subjective opinion)	Yes (Mathematical Formula)
Contradiction / Conflict Detection	No	No	Weak (Fails on long docs)	Yes (Lexical + Semantic Logic)
Traceability Matrix Generation	Manual linking	No	No	Yes (Automated Bidirectional)
Change Impact Blast Radius	No (Manual review)	No	No	Yes (Traverses Modules/Tasks/Tests)

Key contributions implemented in the ReqPilot AI platform include:

• Implementation of an end-to-end full-stack platform uniting React 19, TypeScript, Tailwind CSS, FastAPI, and SQLAlchemy into a cohesive, responsive software engineering workspace.

• Formulation and implementation of a 7-dimensional mathematical quality audit algorithm providing fine-grained score breakdowns across completeness, clarity, atomicity, consistency, testability, feasibility, and traceability.

• Design of an automated clarification loop that translates detected ambiguities and business rule contradictions into structured multiple-choice questions for human stakeholder resolution.

• Creation of a graph-based architectural synthesis and DAG task decomposition engine that automatically maps requirements to microservice modules, estimates story points, and computes sprint schedules.

• Implementation of an automated change impact simulator that tracks requirement mutations across downstream modules, development tasks, test cases, and delivery milestones, preventing sprint regression.

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CHAPTER 3

PROPOSED SYSTEM AND REQUIREMENTS

3.1 PROPOSED SYSTEM

ReqPilot AI is structured as an interactive, intelligent web platform that guides software teams from initial requirement ingestion to sprint planning and change management. From the user's perspective, the application provides a project-centric workspace. Upon creating a project, the user uploads raw software requirement specifications in PDF, DOCX, or TXT format. The platform processes the document through an automated pipeline, presenting extracted requirements in an interactive workspace equipped with rich analytical telemetry, health scores, quality breakdowns, and architectural graphs [18].

Users interact with the system through dedicated functional tabs: Overview, Sources, Requirements, Issues, Clarifications, Architecture, Planning, Traceability, Change Impact, and Reports. High-risk issues, such as contradictory cancellation rules or ambiguous latency adjectives, are flagged visually with severity indicators. Users can resolve issues, approve clarified specifications, adjust architectural modules on an interactive canvas, inspect sprint backlogs, and simulate proposed requirement modifications before committing changes to the database.

3.2 MAJOR FEATURES

The platform architecture organizes system functionality into a clear hierarchy: nine core functional capability domains, ten interactive frontend workspace tabs (Overview, Sources, Requirements, Issues, Clarifications, Architecture, Planning, Traceability, Change Impact, and Reports), and fifteen specialized cooperative backend agents coordinated across an eight-step execution pipeline:

1. Project & Ingestion Manager: Manages project metadata, domains, and health telemetry; ingests multi-format documents, performing OCR, structural normalization, and chunking with source line preservation.

2. Requirement Extraction & Classification Engine: Extracts structured requirements into functional and non-functional categories, isolating Actor, Action, Target Object, Preconditions, and Expected Outcome.

3. 7-Dimensional Quality Audit System: Computes overall requirement quality scores (0100%) and granular breakdowns across completeness, clarity, atomicity, consistency, testability, feasibility, and traceability.

4. Ambiguity & Conflict Detection Hub: Scans specifications against heuristic lexicons of vague terms and cross-references requirement pairs to detect logical business contradictions.

5. Human-in-the-Loop Clarification Workspace: Generates contextual multiple-choice questions with engineering rationales, allowing product owners to clarify requirements and update approved versions.

6. Architecture Synthesis Engine: Decomposes approved requirements into microservice modules, API gateways, database persistence layers, and communication relations, assigning visual coordinates.

7. Sprint Planning & DAG Scheduling Engine: Generates development tasks with estimated hours and story points, maps tasks to team members based on skill proficiencies, and validates dependency DAGs.

8. Test Case & Traceability Generator: Synthesizes integration and performance test cases with concrete preconditions, execution steps, and assertions, maintaining end-to-end bidirectional traceability.

9. Change Impact Blast-Radius Simulator: Models the downstream ripple effect of requirement revisions, identifying impacted modules, tasks, tests, risk severity, and schedule delays.

3.3 FUNCTIONAL REQUIREMENTS

The functional requirements implemented by the ReqPilot AI platform are formally cataloged in Table 3.1 with unique tracking identifiers.

Table 3.1 Functional requirements of the ReqPilot AI platform

Requirement ID	Module / Feature	Description	Verification Method
FR-01	Project Management	System shall allow users to create, view, update, and monitor project health telemetry.	Automated API & UI Test

Requirement ID	Module / Feature	Description	Verification Method
FR-02	Document Ingestion	System shall upload PDF, DOCX, and TXT files, extracting text into indexed semantic chunks.	Upload & Chunking Validation
FR-03	Requirement Extraction	System shall extract structured requirements identifying Actor, Action, Target, and Type.	Extraction Benchmark Test
FR-04	7-D Quality Scoring	System shall compute mathematical quality scores across 7 dimensions using defined weights.	Unit Test Calculation
FR-05	Ambiguity Detection	System shall detect subjective, non-verifiable terms and assign severity levels.	Lexicon Match Assertion
FR-06	Conflict Identification	System shall identify cross-requirement contradictions with high-confidence explanations.	Pairwise Conflict Test
FR-07	Clarification Workflow	System shall generate multiple-choice questions and update requirement versions upon answer.	Clarification API Test
FR-08	Architecture Synthesis	System shall synthesize microservice modules, API routes, and inter-module relations.	Graph Generation Assertion
FR-09	Task & Sprint Planning	System shall decompose requirements into tasks, estimate effort, and construct valid DAGs.	NetworkX DAG Cycle Test
FR-10	Change Impact Simulation	System shall calculate blast-radius items and schedule impact for modified requirements.	Impact Simulation Test

3.4 NON-FUNCTIONAL REQUIREMENTS

The non-functional requirements (NFRs) govern the operational qualities, architectural constraints, and usability standards of the system:

1. Performance Target (NFR-01): Specified operational performance targets mandate that the frontend dashboard renders analytical tabs within 300 ms, backend API endpoints respond within a targeted 250 ms p95 latency under standard local workloads, and document chunking with heuristic extraction for a 15-requirement document targets completion within

2.5 seconds. Measured end-to-end multi-agent pipeline durations range between 33 and 46 seconds on full multi-page benchmark specifications when external cloud LLM reasoning and rate-limited slot cooldowns are active.

2. Security (NFR-02): All API communications shall utilize HTTP/REST protocols with JSON payloads. Authentication shall be enforced via cryptographically signed JSON Web Tokens (JWT) using the HS256 algorithm with passwords salted and hashed using bcrypt (12 rounds). Input sanitization regex-filters known prompt injection phrases (e.g., 'ignore previous instructions') from untrusted document text prior to LLM submission, mitigating direct prompt override attempts.

3. Usability (NFR-03): The interface shall adhere to modern responsive web design standards utilizing Tailwind CSS, maintaining accessibility contrast ratios (> 4.5:1), and providing visual toast notifications for all asynchronous network operations.

4. Reliability (NFR-04): The system shall implement robust graceful degradation. If external cloud LLM APIs (Gemini or Groq) experience rate-limiting, network timeouts, or invalid keys, the platform shall automatically fall back to deterministic regex and heuristic agents without raising unhandled 500 errors.

5. Scalability & Maintainability (NFR-05): The backend shall utilize asynchronous ASGI execution (FastAPI + Uvicorn) with non-blocking SQLite database queries via aiosqlite, structured to support seamless migration to enterprise PostgreSQL clusters.

3.5 USER INTERACTION

Users interact with the platform through a unified, intuitive React application structured around dedicated workspaces. When the user triggers the Multi-Agent Analysis workflow, an interactive Analysis Progress Modal renders real-time telemetry indicating the exact execution stage (Step 1/7 Ingestion, Step 2/7 Extraction, Step 3/7 Quality Auditing, Step 4/7 Ambiguity/Conflict Detection, Step 5/7 Architecture Generation, Step 6/7 Task Decomposition, and Step 7/7 Traceability Linking).

In the Requirements and Issues workspaces, users can click any requirement card to inspect its 7-D quality breakdown radar, view flagged ambiguity terms, or open clarification dialogs. In the Clarifications tab, users are presented with targeted questions where selecting an option (e.g., establishing a 2-hour tiered cancellation policy) immediately refines the underlying requirement, archives the previous version, and triggers automatic re-scoring. In

the Architecture tab, modules and relation arrows are rendered with technology stacks and API endpoints. In the Change Impact tab, users select a target requirement, input a proposed modification, and trigger immediate blast-radius calculation, viewing affected modules, tasks, and tests before committing changes.

3.6 USE CASES

Figure 3.1 illustrates the core use case diagram of ReqPilot AI, capturing interactions between software engineering stakeholders and the platform boundary.

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Figure 3.1 Use case diagram of the ReqPilot AI platform

Table 3.2 details the primary use case: Ingesting a specification, resolving detected conflicts, and generating a validated project plan.

Table 3.2 Primary use case specification: UC-01 Requirements Elicitation & Analysis

Use Case ID & Name	**UC-01: End-to-End Requirements Ingestion, Conflict Resolution & Planning**
Primary Actors	Requirements Engineer, Product Owner, Project Manager
Preconditions	User is authenticated and has created an active project container.
Main Success Scenario (Flow)	1. User uploads 'SMART CAMPUS MANAGEMENT SYSTEM.docx'. 2. System extracts 9 structured requirements and computes 7-D quality scores. 3. Conflict Agent flags contradiction between FR-006 (unrestricted) and FR-007 (24h cutoff). 4. User answers Clarification Question, selecting the 2-Hour Tiered Cutoff option. 5. System updates FR-006 to Version 2, archives Version 1, and recalculates health. 6. System synthesizes 7 architecture modules, 5 sprint tasks, and 3 test cases. 7. Traceability matrix is established and engineering report is generated.
Alternative / Exception Flows	3a. External LLM API key is invalid or offline: System automatically engages local deterministic heuristic extraction without interruption. 4a. User dismisses issue: Requirement remains at current quality score with audit log.
Postconditions	All approved requirements, architecture relations, tasks, and test cases persisted in database.

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CHAPTER 4

SYSTEM DESIGN AND ARCHITECTURE

4.1 OVERALL SYSTEM ARCHITECTURE

The system architecture of ReqPilot AI is organized into a clean, decoupled five-tier layered architecture: (1) Client Presentation Tier, (2) Edge & Gateway Tier, (3) Application & API Tier, (4) Multi-Agent AI Cognitive Tier, and (5) Persistence & Infrastructure Tier [19]. Figure 4.1 provides the complete architectural diagram representing the implemented technologies.

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Figure 4.1 High-level multi-tier system architecture of the ReqPilot AI platform

The execution flow operates as follows: The user interacts with the React 19 single-page application. High-level actions (such as document upload or analysis triggers) are dispatched as asynchronous HTTP REST requests to the FastAPI backend. FastAPI routes validate incoming payloads using Pydantic V2 schemas. The Workflow Orchestrator coordinates the fifteen specialized agents, which query LLM adapters (Gemini 1.5 Flash, Groq LLaMA-3.3-70B, or local heuristics) and perform topological graph computations. All entities, versions, issues, and linkages are persisted into SQLite/PostgreSQL through SQLAlchemy 2.0 Async ORM under ACID transactions.

4.2 FRONTEND DESIGN

The frontend architecture is implemented in TypeScript utilizing React 19, Vite, and Tailwind CSS. The application is structured into four primary component hierarchies:

1. Core Application Shell: Handled by App.tsx, Navbar.tsx, and ProjectHeader.tsx, providing global navigation, project switching, health score telemetry, and responsive layouts.

2. State Management & Context: The centralized AppContext.tsx maintains global application state, including active projects, project statistics, active navigation tabs, API provider selection (Gemini vs. Groq), analysis progress modal state, and auto-dismissing toast notifications.

3. Workspace Tab Views: Ten specialized view components located in src/components/workspace/: OverviewTab.tsx, SourcesTab.tsx, RequirementsTab.tsx, IssuesTab.tsx, ClarificationsTab.tsx, ArchitectureTab.tsx, PlanningTab.tsx, TraceabilityTab.tsx, ChangeImpactTab.tsx, and ReportsTab.tsx.

4. Service Abstraction: The api.ts service encapsulates all network interactions using native asynchronous fetch with strong TypeScript type assertions across all backend schemas.

4.3 BACKEND DESIGN

The backend service is engineered using Python 3.11 and the FastAPI framework, running on the asynchronous Uvicorn ASGI server. The application is structured around high-cohesion, low-coupling design principles. The backend codebase is partitioned into three core directories: app/api/ containing fourteen modular APIRouter controllers, app/agents/ containing the multi-agent cognitive services, and app/models.py defining the SQLAlchemy database schema [20].

Table 4.1 summarizes the authoritative API endpoint catalog implemented in the backend.

Table 4.1 Backend API endpoint summary and route specification

Method	Endpoint Route	Controller File	Functional Purpose
POST	/api/auth/login	auth.py	Authenticate user and issue HS256 JWT access token
GET	/api/projects	projects.py	Fetch list of all projects and health metrics
POST	/api/projects/{id}/documents/upload	documents.py	Upload and chunk specification documents
POST	/api/projects/{id}/analyze	workflow.py	Trigger full 7-step multi-agent analysis pipeline
GET	/api/projects/{id}/requirements	requirements.py	Fetch all functional and non-functional requirements
GET	/api/projects/{id}/issues	issues.py	Retrieve flagged ambiguities, conflicts, and duplicates
POST	/api/clarifications/{id}/answer	clarifications.py	Submit clarification answer and refine requirement

Method	Endpoint Route	Controller File	Functional Purpose
GET	/api/projects/{id}/architecture	architecture.py	Fetch synthesized modules, tech stacks, and relations
GET	/api/projects/{id}/tasks	planning.py	Retrieve decomposed agile tasks and sprint allocations
POST	/api/projects/{id}/ change-impact/simulate	change_impact.py	Simulate blast radius of proposed requirement modification

4.4 DATABASE DESIGN

ReqPilot AI implements an enterprise relational database model managed via SQLAlchemy 2.0 Async ORM with SQLite (aiosqlite) for lightweight deployment and direct compatibility with PostgreSQL 16. The database schema encompasses twenty-three tables organized into three logical functional zones: (1) Core Project & Requirements Zone, (2) Planning & Architecture Zone, and (3) Audit & Change Management Zone.

Figure 4.3 illustrates the entity-relationship (ER) structure connecting the core entities.

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Figure 4.3 Conceptual Entity-Relationship (ER) diagram of core ReqPilot AI database entities

Table 4.2 documents the primary fields, data types, and foreign key relationships of selected core database entities. The active persistence layer implements a comprehensive relational schema comprising twenty-three tables managed via SQLAlchemy 2.0 Async ORM with SQLite (aiosqlite) and PostgreSQL compatibility: users, projects, documents, document_chunks, requirements, requirement_versions, requirement_issues, clarification_questions, architecture_modules, architecture_relations, epics, features, tasks, task_dependencies, sprints, team_members, impact_items, change_requests, test_cases, traceability_links, reports, audit_logs, and ai_workflows.

Table 4.2 Selected core database entity specifications and relational constraints

Table Name	Primary Key	Key Attributes & Types	Foreign Keys & Relationships
projects	id (UUID)	title (str), domain (str), health_score (float)	1:N with requirements, documents, modules

Table Name	Primary Key	Key Attributes & Types	Foreign Keys & Relationships
documents	id (UUID)	filename (str), file_type (str), status (str)	FK: project_id -> projects.id
requirements	id (str, e.g. FR-001)	req_type (str), summary (str), quality_score (float)	FK: project_id -> projects.id; 1:N with issues
requirement_issues	id (UUID)	issue_type (str), severity (str), detected_phrase (str)	FK: requirement_id -> requirements.id
clarification_questions	id (UUID)	question (str), suggested_options (JSON), status (str)	FK: requirement_id -> requirements.id
architecture_modules	id (str, MOD-AUTH)	name (str), module_type (str), tech_stack (str)	FK: project_id -> projects.id; 1:N with tasks
tasks	id (str, TASK-001)	title (str), estimated_hours (float), story_points (int)	FK: requirement_id -> requirements.id, module_id

4.5 MULTI-AGENT AI SYSTEM DESIGN

Rather than deploying a generic conversational chatbot that generates unstructured prose, ReqPilot AI is architected around a specialized, multi-agent cognitive architecture. The platform orchestrates fifteen cooperative agents coordinated by the WorkflowOrchestrator:

1. Ingestion Agent: Performs structural parsing, regex line splitting, and chunk creation.

2. Extraction Agent: Employs schema-constrained LLM adapters to extract structured requirements with evidence anchors, falling back to regex token matching.

3. 7-D Quality Agent: Implements mathematical scoring across completeness, clarity, atomicity, consistency, testability, feasibility, and traceability.

4. Ambiguity Agent: Detects vague linguistic phrases using an extensive heuristic dictionary of non-testable terms.

5. Conflict Agent: Performs pairwise logical analysis to detect contradictory business constraints.

6. Duplicate Agent: Computes TF-IDF and character trigram cosine similarity to detect duplicate requirements (> 0.65 threshold).

7. Clarification Agent: Formulates contextual multiple-choice questions with engineering rationales.

8. Architecture Agent: Synthesizes modular microservices, gateways, and communication relations.

9. Task Decomposition Agent: Breaks requirements into development tasks with acceptance criteria.

10. Estimation Agent: Computes effort hours and Fibonacci story points based on requirement complexity.

11. Resource Matching Agent: Maps tasks to team members based on skill match scores.

12. Scheduling Agent: Models dependencies as a Directed Acyclic Graph (DAG) and computes topological sprint order.

13. Traceability Agent: Synthesizes bidirectional links across source, requirement, module, task, and test.

14. Change Impact Agent: Calculates the blast radius and schedule risk of requirement modifications.

15. Report Agent: Consolidates all engineering artifacts into an auditable executive report.

4.6 DATA FLOW AND WORKFLOW PIPELINE

The operational data flow of ReqPilot AI represents a continuous, closed-loop engineering pipeline. Figure 4.2 illustrates the complete end-to-end processing sequence.

–

Figure 4.2 Data flow and multi-agent workflow pipeline of ReqPilot AI

–

4.7 DYNAMIC CAPABILITY AND MODULE DISCOVERY

A foundational architectural advancement introduced in ReqPilot AI is genuine, project-agnostic capability discovery. In traditional requirement management tools and early baseline architectures, requirements are coerced into a rigid, predefined taxonomy (e.g., forcing all project entities into fixed modules such as MOD-BOOKING, MOD-AUTH, or

MOD-DATA). When deployed against novel industrial domains—such as Precision Agriculture, Autonomous Robotics, Telecommunications, or Complex Financial Instruments—such rigid taxonomies fail catastrophically by stripping domain semantics, producing inaccurate microservice boundaries, and misrepresenting business rules.

To achieve genuine project generalization, ReqPilot AI implements an evidence-grounded dynamic synthesis engine within ArchitectureAgent and Taxonomy. Rather than coercing requirement text through broad keyword tokens (e.g., regex matching 'order' to MOD-BOOKING), the pipeline empowers the cognitive architecture tier to synthesize domain-authentic capabilities directly from the client document. In an unfamiliar precision agriculture specification, for example, the system autonomously derives evidence-backed modules such as MOD-DRONE-NAVIGATION, MOD-IMAGING, MOD-SENSOR-INGESTION, and MOD-ACTUATION. These dynamic capabilities survive downstream task decomposition, NetworkX DAG scheduling, and traceability matrix construction intact, ensuring that the synthesized architecture accurately reflects the domain without artificial taxonomy distortion.

CHAPTER 5

IMPLEMENTATION

5.1 DEVELOPMENT ENVIRONMENT

The implementation of ReqPilot AI utilizes robust open-source technologies across the complete software stack. Table 5.1 outlines the verified development environment, runtime versions, libraries, and build tools.

Table 5.1 Development environment, software versions, and dependencies

Layer / Domain	Technology & Version	Purpose & Architectural Role
Frontend Framework	React 19.2.8 + TypeScript 6.0.2	Core component-driven user interface architecture
Build Tool & Bundler	Vite 6.2.0 + @tailwindcss/vite	Ultra-fast HMR development and optimized production build
Styling & Icons	Tailwind CSS 4.3.3 + Lucide React 1.41.0	Utility-first responsive styling and SVG iconography

Layer / Domain	Technology & Version	Purpose & Architectural Role
Backend Runtime	Python 3.11.9 + FastAPI 0.115.0	High-concurrency asynchronous ASGI REST API framework
Database & ORM	SQLAlchemy 2.0.30 + aiosqlite / SQLite	Asynchronous relational persistence and ACID transactions
Graph & ML Analytics	NetworkX 3.3 + Scikit-Learn 1.5.0	DAG topological sorting and lexical similarity clustering
LLM Integration	Google Gemini 1.5 Flash / Groq LLaMA-3.3	Schema-constrained generative reasoning and question synthesis

5.2 FRONTEND IMPLEMENTATION

The frontend implementation centers on modular, state-driven React components. AppContext.tsx serves as the single source of truth, managing active project context, analysis execution state, and API credentials. TabNavigation.tsx provides high-visibility tab navigation across all ten workspaces. When an action is initiated (such as answering a clarification question or simulating a change), the component dispatches an asynchronous call via api.ts and immediately updates the local state upon receiving a 200 OK response.

5.3 BACKEND IMPLEMENTATION

The backend is implemented as a modular FastAPI application initialized in app/main.py. CORS middleware is configured to permit secure cross-origin communication from the Vite development server. Each API controller in app/api/ relies on FastAPI dependency injection to obtain asynchronous database sessions (AsyncSessionLocal). Request payloads are strictly validated against Pydantic models defined in app/schemas.py, preventing malformed JSON from reaching database tables.

5.4 DATABASE IMPLEMENTATION

The persistence layer utilizes SQLAlchemy 2.0 Async ORM. Database tables are defined declaratively in app/models.py using mapped_column and relationship constructs with cascade deletions. Asynchronous database transactions ensure that complex multi-table inserts (e.g., seeding a project with requirements, modules, tasks, and issues) execute atomically. The database file reqpilot.db is managed via asynchronous SQLite, providing zero-configuration portability while maintaining full compatibility with enterprise PostgreSQL engines.

5.5 MULTI-AGENT AI IMPLEMENTATION

The multi-agent cognitive tier is implemented in app/agents/. The central LLMAdapter class wraps HTTP requests to Google Generative AI (Gemini 1.5 Flash), Groq (LLaMA-3.3-70B), or OpenAI. A critical security feature implemented in LLMAdapter is input sanitization (sanitize_untrusted_input), which regex-strips adversarial prompt injection phrases (e.g., 'ignore previous instructions') from uploaded document text before invoking LLM endpoints. Furthermore, when API keys are absent, all agents smoothly fall back to internal rule-based heuristic engines, ensuring high operational availability without unhandled API exception crashes.

5.6 FULL STACK INTEGRATION

The end-to-end integration across frontend, backend, database, and AI services is orchestrated through clean RESTful interfaces. When a user clicks 'Run Analysis' in the frontend:

1. Frontend sends POST /api/projects/{id}/analyze with provider selection.

2. workflow.py loads uploaded document chunks from SQLite.

3. WorkflowOrchestrator invokes the Extraction, Quality, Ambiguity, Conflict, Architecture, Task, and Traceability agents sequentially.

4. Generated records are persisted into SQLite under an ACID transaction.

5. Frontend receives the consolidated execution response, closes the progress modal, and refreshes project health telemetry.

5.7 IMPORTANT CODE SNIPPETS

The following code listings demonstrate the core algorithms implemented in ReqPilot AI.

Listing 5.1 Mathematical 7-Dimensional Quality Scoring Implementation (quality_agent.py)

```
class QualityAgent:
    """Evaluates requirements across 7 transparent dimensions aligned with ISO 29148."""
    def evaluate_requirement(self, req: Dict[str, Any], all_reqs: List[Dict[str, Any]] = None) ->
Dict[str, Any]:
        breakdown = {
            "completeness": self._eval_completeness(req),
            "clarity": self._eval_clarity(req),
            "atomicity": self._eval_atomicity(req),
            "consistency": self._eval_consistency(req, all_reqs or []),
```

```
        "testability": self._eval_testability(req),
        "feasibility": self._eval_feasibility(req),
        "traceability": self._eval_traceability(req),
    }
    weights = settings.QUALITY_WEIGHTS
    score = sum(breakdown[dim] * weights[dim] for dim in weights)
    return {"quality_score": round(score, 1), "breakdown": breakdown}
```

Listing 5.2 NetworkX DAG Task Scheduling with Fallback Handling (scheduling_agent.py excerpt)

```
class SchedulingAgent:
    """Models task dependencies as a DAG and computes topological execution order
(excerpt)."""
    def schedule_tasks(self, tasks: List[Dict[str, Any]], dependencies: List[Dict[str, Any]])
-> Dict[str, Any]:
        G = nx.DiGraph()
        for t in tasks:
            G.add_node(t["id"], title=t.get("title", ""), points=t.get("story_points", 3))
        for d in dependencies:
            G.add_edge(d["depends_on_task_id"], d["task_id"])

        is_dag = nx.is_directed_acyclic_graph(G)
        if is_dag:
            # Valid acyclic dependency graph: compute optimal topological sprint execution
order
            order = list(nx.topological_sort(G))
        else:
            # Dependency cycle detected: returns original task sequence as a graceful
fallback,
            # flagged with is_valid_dag = False to alert engineering authorities of dependency
deadlocks.
            order = [t["id"] for t in tasks]
        return {"is_valid_dag": is_dag, "execution_order": order}
```

Listing 5.3 Change Impact Blast-Radius and Heuristic Risk Assessment (change_impact_agent.py excerpt)

```
class ChangeImpactAgent:
    """Evidence-based change impact analysis traversing tasks, modules, and tests
(excerpt)."""
    def analyze_change(self, target_req_id: str, new_summary: str, new_description: str,
                       all_requirements: List[Dict], all_modules: List[Dict],
                       all_tasks: List[Dict], all_tests: List[Dict]) -> Dict[str, Any]:
        # 1. Directly affected tasks linked via foreign key requirement_id
        affected_tasks = [t for t in all_tasks if t.get("requirement_id") == target_req_id]
        # 2. Directly affected tests linked via requirement_id
        affected_tests = [t for t in all_tests if t.get("requirement_id") == target_req_id]
        # 3. Architecture modules mapped through affected tasks
```

```
        module_ids = {t.get("module_id") for t in affected_tasks if t.get("module_id")}
        # 4. Related requirements identified via token similarity overlap (threshold >= 0.30)
        related_reqs = self._find_related_requirements(target_req_id, new_summary,
new_description, all_requirements)
        # 5. Illustrative risk assessment heuristic based on artifact count and task
complexity
        total_affected = len(affected_tasks) + len(affected_tests) + len(module_ids) +
len(related_reqs)
        high_complexity = sum(1 for t in affected_tasks if str(t.get("complexity") or
"").lower() == "high")
        risk = "critical" if (high_complexity >= 2 or total_affected >= 10) else ("high" if
total_affected >= 5 else "medium")
        schedule = "High" if risk == "critical" else ("Medium" if risk == "high" else "Low")
        return {"target_requirement_id": target_req_id, "risk_level": risk, "schedule_impact":
schedule,
                "summary_stats": {"affected_modules": len(module_ids), "affected_tasks":
len(affected_tasks),
                                  "affected_tests": len(affected_tests), "total_artifacts":
total_affected}}
```

—

5.8 PROJECT-AGNOSTIC GENERALIZATION AND DYNAMIC MODULE PRESERVATION

During production pipeline engineering and cross-domain remediation sprints, two subtle yet pervasive coercion defects were identified and eliminated from the production codebase. First, in backend/app/agents/taxonomy.py, the normalize_module_id function previously performed broad regex token searches across input slugs. As a result, compound dynamic module identifiers such as MOD-ORDER-EXECUTION or MOD-INPATIENT-ORDER were inadvertently coerced into fixed taxonomy modules (MOD-BOOKING) whenever common substring tokens matched. This was resolved by migrating to exact alias and prefix matching on clean_unprefixed strings while consulting canonical entity aliases only for verified general terms.

Second, in backend/app/agents/task_agent.py, candidate module validation was re-architected. In earlier iterations, the task decomposition validation logic attempted taxonomy normalization before checking whether a proposed module was already present in the architecture candidate set. This caused dynamically synthesized candidate modules to be aggressively overwritten by legacy fallback rules. The resolution inverted this logic: candidate modules proposed by ArchitectureAgent are validated first and preserved with

strict referential integrity. A dedicated automated test suite (backend/tests/test_dynamic_modules.py) validates unknown module preservation, invalid ID rejection, duplicate slug resolution, and end-to-end traceability matrix generation across dynamic modules, expanding the test suite to 46 passing unit tests.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 SYSTEM OUTPUT AND DEMONSTRATION WORKSPACE

ReqPilot AI was executed and verified against representative full-stack software requirements specifications. The application ingests raw stakeholder documents in PDF, Microsoft Word (.docx), or plaintext (.txt) formats, performs automated structural cleaning and semantic chunking, and dynamically instantiates an enterprise project workspace. Figure 6.1 illustrates the responsive user interface dashboard, showcasing project health metrics, 7-dimensional quality breakdowns, architecture canvas nodes, and sprint backlog allocations.

–

Figure 6.1 User interface of the ReqPilot AI platform workspace

6.2 MULTI-AGENT COGNITIVE INTERACTION TRACES

The cognitive intelligence of ReqPilot AI was benchmarked across representative elicitation workflows. The following authentic interaction traces demonstrate the platform's multi-agent coordination during ambiguity detection, clarification question synthesis, and cross-document business rule conflict resolution:

--- Trace 1: Ambiguity Detection and Clarification Dialogue ---

User Ingestion: Requirement NFR-002: 'The tutor booking search must be fast and responsive.'

Multi-Agent Processing: AmbiguityAgent scans text and detects non-measurable adjective 'fast' (confidence 0.92). QualityAgent penalizes Clarity (40) and Testability (50), yielding an overall quality score of 62.0. ClarificationAgent formulates targeted multiple-choice question: 'What maximum response time should be targeted for tutor search under peak load?' Options: ['p95 < 250ms for search; p99 < 500ms for bookings', '< 1 second for all endpoints', '< 2 seconds under peak load'].

System Response: User selects Option 1. NFR-002 is refined to 'Tutor search response time shall maintain p95 < 250ms under peak load of 1,000 concurrent queries'. Quality score is recalculated and elevated to 94.0.

--- Trace 2: Cross-Requirement Conflict Resolution ---

User Ingestion: Document contains FR-006 ('Students may cancel bookings at any time without restriction') and FR-007 ('Bookings cannot be cancelled within 24 hours of session start').

Multi-Agent Processing: ConflictAgent detects mutual contradiction (confidence 0.96). Conflict issue is flagged in Issues workspace with high severity. Clarification question generated with tiered cancellation policy options.

System Response: Stakeholder selects Tiered 2-Hour Cutoff. FR-006 is updated to Version 2: 'Full refund up to 24h prior, 50% refund between 2-24h, no refund under 2h'. Contradiction issue is closed as resolved.

6.3 FUNCTIONAL TESTING AND AUTOMATED REGRESSION SUITE

Automated unit, integration, and regression test suites were executed across the platform using pytest 8.3 and httpx. The test harness spans 46 comprehensive test cases covering multi-agent cognitive services (QualityAgent, AmbiguityAgent, ConflictAgent, DuplicateAgent, SchedulingAgent, ChangeImpactAgent), FastAPI REST endpoints, and dynamic module preservation (test_agents.py, test_workflow.py, test_classification_remediation.py, and test_dynamic_modules.py). All 46 unit and regression tests executed with zero failures (46 passed, 16 deprecation warnings in 10.09s on 2026-10-14). Table 6.1 details the functional test execution matrix across core platform components.

Table 6.1 Functional test execution matrix and verification results

Test ID	Target Component	Input / Precondition	Observed Output	Status
TC-01	QualityAgent	Requirement with full actor, outcome, and criteria	Quality Score = 95.5, 7-D breakdown verified	**PASS**
TC-02	AmbiguityAgent	Requirement containing terms 'fast' and 'simple'	Flagged 1 issue with detected phrase 'Fast'	**PASS**
TC-03	ConflictAgent	Opposing cancellation policies (FR-006 vs FR-007)	Detected high-severity contradiction issue	**PASS**
TC-04	DuplicateAgent	Identical registration requirements (0.65 threshold)	Detected 1 duplicate pair with similarity score	**PASS**
TC-05	SchedulingAgent	2 tasks with dependency link (TASK-001 -> TASK-002)	is_valid_dag = True, topological order verified	**PASS**

Test ID	Target Component	Input / Precondition	Observed Output	Status
TC-06	ChangeImpactAgent	Modified FR-006 cancellation cutoff requirement	Affected tasks: 1, modules: 1, tests: 1	**PASS**

6.4 OFFICIAL FOUR-DOMAIN BENCHMARK EVALUATION RESULTS

To rigorously evaluate the semantic classification, architectural synthesis, and generalization performance of ReqPilot AI, the production pipeline was evaluated across four diverse industrial benchmark projects using the official evaluation runner (evaluation/multi_project/scripts/run_eval.py) and scored against frozen ground-truth references using the official, cryptographically verified scorer (evaluation/multi_project/scripts/scorer.py, SHA-256 hash: b6c599655cc7c8e8fe8550c8e74c2bd13335ff6ec41f859a0354b3932a3655fb).

Evaluation Scope Disclosure: ReqPilot AI achieved the reported metrics on four specified benchmark projects: Library Management, Banking Management, E-Commerce Platform, and EV Charging. Hospital Management was excluded from this active evaluation suite and safely archived under evaluation/multi_project/archived/hospital_management/ for separate ground-truth review. These results demonstrate performance on the evaluated projects and do not establish universal reliability or performance on unseen domains.

Table 6.2 presents the official quantitative benchmark results across all four retained domains from the verified evaluation snapshot, detailing run identifiers, gold requirement sample counts, matched denominators, classification accuracy, penalized end-to-end accuracy, and macro-F1. Aggregate macro-F1 is calculated as the unweighted arithmetic mean of individual domain macro-F1 scores (0.8710 across the four domains), treating each domain equally regardless of requirement count.

Table 6.2 Official multi-project benchmark evaluation results across the four retained domains

Project Domain	Run Identifier	Gold Reqs	Matched Denom	Correct Class	Accuracy (%)	Macro-F1
EV Charging	9d9a6a23 (remediation)	58	58	56	96.55%	0.9741

Project Domain	Run Identifier	Gold Reqs	Matched Denom	Correct Class	Accuracy (%)	Macro-F1
E-Commerce Platform	a0a716cb (remediation)	36	36	35	97.22%	0.9677
Library Management	221074ed (remediation)	24	24	23	95.83%	0.9604
Banking Management	1616700d (remediation)	25	25	21	84.00%	0.5818
Aggregate (4 Domains)	**Verified Official Suite**	**143**	**143**	**135**	**94.41%**	**0.8710**

Figure 6.2 illustrates the comparative performance across all four benchmark projects in terms of classification accuracy and macro-F1 harmonic scores. Figure 6.3 summarizes the automated verification and unit test pass rates.

–

Figure 6.2 Requirements classification accuracy and macro-F1 across the four retained domains

–

Figure 6.3 Test suite verification pass rate and structural integrity metrics

6.5 CLASSIFICATION ERROR ANALYSIS AND BENCHMARK BOUNDARIES

Across the 143 total evaluated requirements in the active four-domain suite, ReqPilot AI correctly classified 135 requirements, yielding an aggregate accuracy of 94.41% with exactly 8 misclassifications. Rather than indicating catastrophic pipeline failures, an independent investigation reveals that these 8 cases stem from nuanced semantic boundary tensions between dynamic capability discovery and static gold-reference annotations:

1. Banking Management Misclassifications (4 errors): In Banking Management, the model accurately recognized specialized business domains and dynamically proposed LOAN MANAGEMENT for FR-005 ('Customer online loan application') and FR-018 ('Loan officer review and approvals'). However, the benchmark gold reference expected generic MOD-BOOKING. Similarly, for FR-007 ('Debit card activation, freezing, limits'), the model synthesized DEBIT CARD MANAGEMENT, whereas the gold reference expected MOD-PROFILE. In NFR-003 ('Customer financial record protection'), the model assigned MOD-DATA, whereas gold expected MOD-AUTHENTICATION. These cases highlight the

tension between authentic microservice capability synthesis and fixed-category benchmark scoring.

2. EV Charging Misclassifications (2 errors): Across 58 requirements, 56 were correctly classified (96.55% accuracy). The two edge cases involved energy load balancing and automated grid throttle policies, which were assigned to MOD-STATION-OPS rather than MOD-PAYMENT or MOD-ANALYTICS.

3. E-Commerce Platform Misclassification (1 error): Across 36 requirements, 35 were correctly classified (97.22% accuracy). The single discrepancy involved an automated order cancellation webhook event, which was allocated to MOD-ORDER rather than MOD-NOTIFICATION.

4. Library Management Misclassification (1 error): Across 24 requirements, 23 were correctly classified (95.83% accuracy). The single difference involved an overdue fine payment alert, which was routed to MOD-NOTIFICATION rather than MOD-CIRCULATION.

6.6 RETIREMENT AND ARCHIVAL OF THE HOSPITAL MANAGEMENT BENCHMARK

Hospital Management System was formally retired from ReqPilot AI's active evaluation suite and archived under evaluation/multi_project/archived/hospital_management/. A rigorous ground-truth audit of its 124 requirements revealed severe internal annotation defects. Over 48 requirements contained ambiguous or overlapping module assignments in the original gold reference (e.g., audit logging assigned to MOD-DATA rather than MOD-AUDIT, search operations assigned to arbitrary clinical modules rather than MOD-SEARCH, and clinical workflows split inconsistently). Furthermore, earlier high scores had been achieved through broad taxonomy regex coercion, which artificially masked annotation defects. When dynamic module discovery was enabled, the model stopped forcing inputs into the fixed taxonomy, correctly exposing the underlying ground-truth flaws (50.00% accuracy). To preserve scientific rigor, all historical run artifacts and audit files were preserved and archived rather than silently deleted, ensuring complete transparency and reproducibility while removing distorted data from the active evaluation suite.

6.7 GENERALIZATION ASSESSMENT ACROSS UNFAMILIAR DOMAINS

To evaluate generalization on domains outside the standard benchmark set, ReqPilot AI was tested against a held-out synthetic specification for Precision Agriculture & Autonomous Drone Telemetry. The architecture synthesis agent successfully discovered five evidence-grounded capabilities: MOD-DRONE-NAVIGATION, MOD-IMAGING, MOD-SENSOR-INGESTION, MOD-ACTUATION, and MOD-ENV-MONITOR. All dynamic modules survived downstream task decomposition, topological scheduling, and traceability linking without coercion. While these results confirm that ReqPilot AI's pipeline architecture is project-agnostic, universal generalization cannot be claimed based on four benchmark projects alone; formal cross-domain reliability requires continued evaluation across larger, independently curated industrial specifications.

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENT

7.1 CONCLUSION

This project designed, implemented, and verified ReqPilot AI, a full-stack multi-agent platform for intelligent software requirements engineering and project planning. Built using React 19, TypeScript, Vite, Tailwind CSS, FastAPI, and SQLAlchemy Async ORM, the platform delivers a structured digital thread connecting raw stakeholder documents to sprint backlogs and verification test suites [21].

By coordinating specialized cognitive agents under a deterministic workflow orchestrator, ReqPilot AI effectively mitigates common LLM vulnerabilities such as schema drift and ungrounded generation. The system automates document ingestion, 7-dimensional quality auditing, ambiguity and conflict resolution, microservice architecture synthesis, DAG sprint task scheduling, and change-impact blast-radius analysis while keeping human engineering authorities firmly in control.

7.2 MAJOR CONTRIBUTION

The primary contributions realized in this project include:

1. Development of an end-to-end full-stack web application delivering an intuitive workspace for requirements engineers, architects, project managers, and QA professionals.

2. Formulation of a mathematically transparent 7-D quality evaluation framework aligned with ISO/IEC/IEEE 29148 standards.

3. Engineering of an automated change impact blast-radius simulator that dynamically calculates downstream module, task, and schedule repercussions upon requirement mutation.

4. Empirical demonstration of 94.41% aggregate matched classification accuracy (135/143 requirements correct) and 0.8710 average macro-F1 across four active industrial benchmark domains (EV Charging, E-Commerce Platform, Library Management, and Banking Management), supported by 46/46 passing automated unit and regression tests.

7.3 LIMITATIONS

While ReqPilot AI represents a comprehensive full-stack platform, several operational limitations are acknowledged:

1. External Cloud LLM Quotas: Advanced semantic synthesis relies on third-party cloud LLM APIs (Gemini 1.5 Flash or Groq). Under strict quota limitations, the system automatically falls back to deterministic heuristic engines, which have lower semantic nuance.

2. Tabular and Diagram Ingestion: The current ingestion agent processes narrative text from PDF, DOCX, and TXT files. Complex embedded flowchart images or multi-page nested tables within Word documents require manual text extraction.

3. User-Level Multi-Tenancy: The database architecture supports multi-project isolation, but enterprise multi-tenant organization hierarchies (e.g., enterprise single sign-on via SAML/OAuth2) remain to be implemented.

7.4 FUTURE ENHANCEMENT

Future enhancements planned for the evolution of ReqPilot AI include:

1. Real-Time Collaborative Canvas: Integrating WebSockets and Yjs CRDTs to enable simultaneous multi-user editing of architecture diagrams and requirement cards, similar to Figma or Miro.

2. Enterprise Jira and GitHub Sync: Developing bi-directional webhooks that automatically export generated tasks, epics, and story points directly into corporate Jira boards and GitHub Issues repositories.

3. Automated Gherkin & Playwright Generation: Expanding test case generation to produce executable Gherkin syntax ('Given-When-Then') and end-to-end Playwright/Cypress automated test scripts.

4. Local Fine-Tuned SLM Deployment: Fine-tuning a lightweight open-source Small Language Model (e.g., LLaMA-3.2-3B or Mistral-7B) using LoRA for fully air-gapped, zero-cost offline enterprise deployments.

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APPENDIX

APPENDIX A: SYSTEM INSTALLATION AND SETUP GUIDE

ReqPilot AI is structured as a decoupled full-stack application. To configure and execute the platform locally, execute the following steps:

1. Backend Setup (FastAPI & Python 3.11):

```
$ cd fsd_pro/backend
$ python -m venv .venv
$ source .venv/bin/activate  (or .venv\Scripts\activate on Windows)
$ pip install -r requirements.txt
$ cp .env.example .env
$ uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
```

2. Frontend Setup (React 19, TypeScript, Vite):

```
$ cd fsd_pro/frontend
$ npm install
$ npm run dev
```

Vite development server will launch at http://localhost:5173 with proxy configured to backend port 8000.

APPENDIX B: GITHUB REPOSITORY SPECIFICATION

The complete source code, frontend components, backend routers, database schemas, multi-agent AI services, unit test suites, and demonstration datasets are version-controlled in the official repository:

Official Repository URL: [INSERT VERIFIED GITHUB REPOSITORY URL]

Repository Contents Structure:

• /fsd_pro/frontend: Complete React 19 + TypeScript + Tailwind CSS application source code.
• /fsd_pro/backend: FastAPI REST application, SQLAlchemy async models, and API routers.
• /fsd_pro/backend/app/agents: Fifteen specialized multi-agent cognitive services.
• /fsd_pro/backend/tests: Comprehensive automated pytest unit and integration test suites.
• /fsd_pro/data/evaluation: Benchmark evaluation dataset containing ground-truth test cases.
• /fsd_pro/backend/uploads: Sample requirements specification documents (PDF, DOCX, TXT).